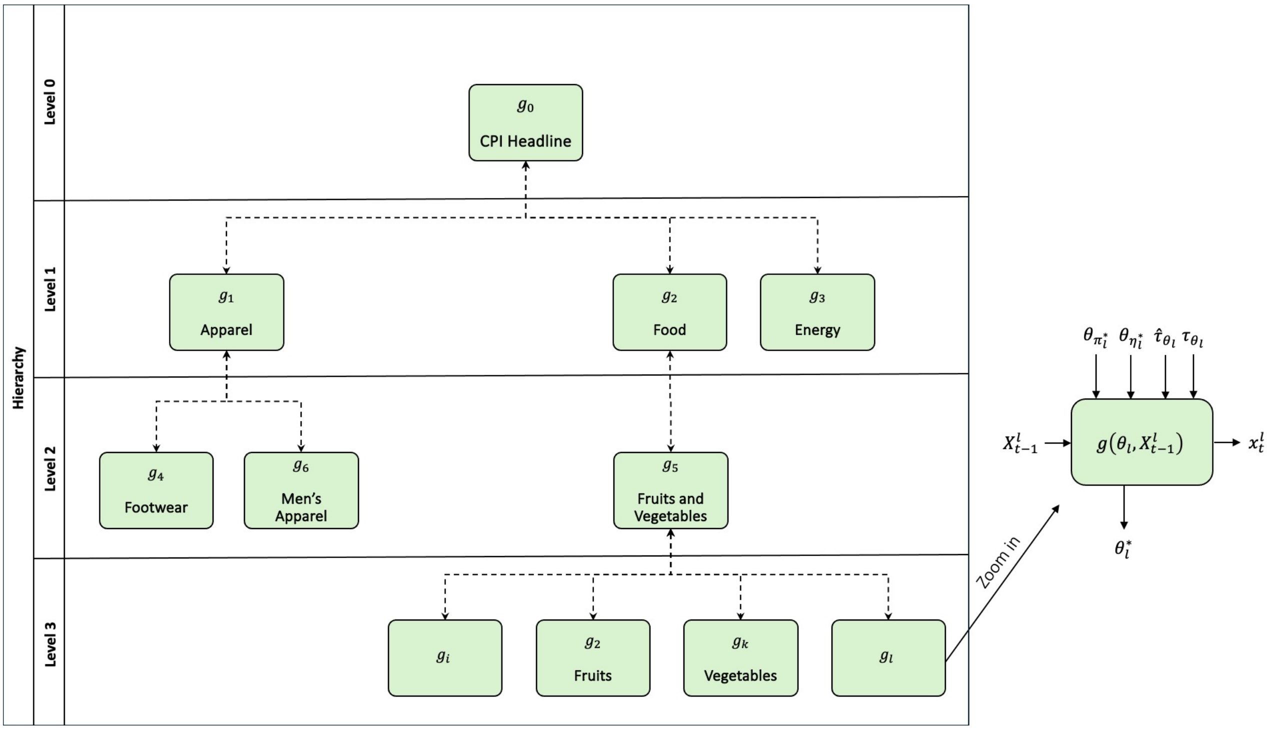
Figure 3.3. *An illustration of our BiHRNN Model.*



4 Dataset

This work is based on monthly CPI data released by the US Bureau of Labor and Statistics (BLS) [1], Statistics Canada [2], and Statistics Norway [3], covering different time periods: the US dataset spans from January 2012 to September 2023, the Norway dataset from January 2009 to May 2023, and the Canada dataset from January 2013 to February 2023. In the following section, we outline the dataset's characteristics and detail our pre-processing methodology. To ensure reproducibility, the final version of the processed data is included within our BiHRNN code.

4.1 *The US Consumer Price Index*

The CPI for each month is released by the BLS a few days into the following month. Price data is gathered from around 24,000 retail and service establishments across 75 urban areas in the US. Additionally, housing and rent information is collected from approximately 50,000 landlords and tenants nationwide. The BLS provides two distinct CPI measurements based on urban demographics:

1. The **CPI-U**, or Consumer Price Index for All Urban Consumers, covers about 93% of the total population. The items included in the CPI, along with their relative weights, are determined based on the Consumer Expenditure Survey, which estimates household spending. These items and weights are updated annually each January.

2. The **CPI-W**, or Consumer Price Index for Urban Wage Earners and Clerical Workers, covers around 29% of the population. This index focuses on households where at least 50% of income is earned through wage-paying or clerical jobs, with at least one household member employed for 70% or more of the year. The CPI-W is commonly used to track changes in benefit costs and inform future contract obligations.

[1] Taken from `www.bls.gov/cpi/overview.htm`.

[2] Taken from `www.statcan.gc.ca/en/start`.

[3] Taken from `www.ssb.no/en`.

In this work, we focus on the CPI-U, as it is widely regarded as the most reliable indicator of the average cost of living in the United States. The CPI-U prices are seasonally adjusted, with updates released in February that reflect price movements from the previous calendar year. It is important to note that, over time, new indexes have been introduced while others have been discontinued, causing shifts in the hierarchical structure of the dataset, which adds complexity to our analysis.

4.1.1 *US CPI Hierarchy*

The CPI-U is structured as an nine-level hierarchy consisting of 350 distinct nodes (indexes). At the top, Level 0 represents the headline CPI, which is the aggregated index of all components. Each index in the hierarchy is assigned a weight between 0 and 100, reflecting its contribution to the headline CPI at Level 0.

Level 1 contains the 8 (not including "All items excluding X" categories) main aggregate categories, or sectors: (1) "Food and Beverages," (2) "Housing," (3) "Apparel," (4) "Transportation," (5) "Medical Care," (6) "Recreation," (7) "Education and Communication," and (8) "Other Goods and Services."

Mid-levels (2-5) include more specific groupings such as "Milk" and "Motor Fuel." The lower levels (6-8) feature finer-grained indexes, such as "Rice," "Child Care and Nursery School," "Legal Services," "Funeral Expenses," and "Parking Fees and Tolls," among others.

4.2 *Canada Consumer Price Index*

The CPI in Canada is measured on a monthly basis by Statistics Canada[4]. The target population for the CPI includes families and individuals living in urban and rural private households, excluding those in communal or institutional settings like prisons or long-term care. The price sample is gathered from various geographic areas, goods, services, and retail outlets to estimate price changes. Outlet selection focuses on high-revenue retailers, with prices mainly collected from retail stores and agencies. The relative importance of items in the CPI basket is derived using data from the Household Final Consumption Expenditure (HFCE) and the Survey of Household Spending (SHS). It should be noted that in some years, category weights for the CPI were unavailable. Therefore, we employed Linear Regression to estimate the missing weights, with the CPI values of the child categories serving as independent variables and the parent category CPI as the dependent variable. We will expand on this further in Section 4.4.1.1.

[4] https://www.statcan.gc.ca/en/start